

```
import pandas as pd
import numpy as np
from sklearn.tree import DecisionTreeClassifier
from sklearn.tree import plot_tree

df=pd.read_csv("/content/drive/MyDrive/summa/Book1.csv",sep=",")
print(df.head())
```

	State	Literacy	Cleanliess	Crime_Rate	Output
0	A	55	88	44	1
1	B	44	56	33	1
2	C	24	33	55	0
3	D	15	22	88	1
4	E	24	33	29	0

```
df.isnull().sum()
```

```
State      0
Literacy   0
Cleanliess 0
Crime_Rate 0
Output     0
dtype: int64
```

```
feature=df[['Literacy','Cleanliess','Crime_Rate']]
target=df['Output']
```

```
DT=DecisionTreeClassifier()
DT.fit(feature,target)
```

```
DecisionTreeClassifier()
```

```
DecisionTreeClassifier(criterion='entropy')
```

```
DecisionTreeClassifier(criterion='entropy')
```

```
DT.score(feature,target)
```

```
1.0
```

```
l=int(input("enter literacy: "))
cl=int(input("enter cleanliness: "))
cr=int(input("enter crime rate: "))
pred=DT.predict([[l,cl,cr]])
if pred==1:
    print("Good State")
else:
    print("Bad State")
```

```
enter literacy: 34
enter cleanliness: 54
```

enter crime rate: 23

Bad State

/usr/local/lib/python3.10/dist-packages/sklearn/base.py:493:

UserWarning: X does not have valid feature names, but

DecisionTreeClassifier was fitted with feature names

warnings.warn(